Karnaugh Map

→ It is a graphical method to simplify complicated boolean functions.

Number of cells:

→ A K-map contains 2^n cells if there are n-variables in the boolean expression or functions.

The K-map method is faster and more efficient than other simplification.

A kmap is a pictorial method used to minimize boolean expression without having to use boolean algebra theorems and equation manipulations.

Two variables K-Map

→ No of cells in k-map = $2^n = 2^2 = 4$

B \ A	0	1
0	$\bar A\bar B$	$\bar A B$
1	$A\bar B$	AB

→ $\bar A = 0$ & $A = 1$

→ $\bar B = 0$ & $B = 1$

Three variables in k-map

→ no of variables = n = 3

→ no of cells = $2^n = 2^3 = 8$

A \ BC	00	01	11	10
0	$\bar A\bar B\bar C$	$\bar A\bar B C$	$\bar A BC$	$\bar A B\bar C$
1	$A\bar B\bar C$	$A\bar B C$	ABC	$AB\bar C$

A kmap can be thought as special version of a truth table.

Four variables K-Map:

→ no of variables = n = 4

→ no of cells in k-map = $2^4 = 16$

AB \ CD	00	01	11	10
00	$\bar A\bar B\bar C\bar D$	$\bar A\bar B\bar C D$	$\bar A\bar B CD$	$\bar A\bar B C\bar D$
01	$\bar A B\bar C\bar D$	$\bar A B\bar C D$	$\bar A BCD$	$\bar A BC\bar D$
11	$AB\bar C\bar D$	$AB\bar C D$	$A\bar B CD$	$ABC\bar D$
10	$A\bar B\bar C\bar D$	$A\bar B\bar C D$	$A\bar B CD$	$A\bar B C\bar D$

Selection of Adjacent Cell in k-map

→ The number of adjacent cells in a k-map selected in the following sequence

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

$F(ABCD) = \bar C + D + B$

Examples:

$F = A\bar{B} + \bar{A}\bar{B} + AB$

A \ B	0	1
0	1	
1	1	1

$= A + \bar{B}$ Ans

→ While reducing the results from the selected the adjacent cells. The repeated variables will remain intact.

→ Where, the difference variable be omitted.

Representing Boolean function in Terms of min-Terms:

$F = m_0 + m_1 + m_2 + \dots + m_{15}$

$F = \sum m(0, 1, 2, \dots, 15)$
(Short hand notation)

K-maps

=> Completely Specified function
=> Incompletely specified function
=> Minimization of minterms
=> Minimization of maxterms (POS)

Notation:

=> $\sum (0, 4, 6, 7)$ Minterms
=> $\pi (1, 2, 3, 5)$ Maxterms

SOP Minimization

Prime implicant:
Are found adjacent cells of K-map containing minterms.

Essential implicant:
Prime implicant that covers at least one minterm which is not been covered by other prime implicants.

Redundant implicant:

Prime implicant which is not essential is called redundant implicant.

Cyclic prime implicants

if we have 6 prime implicant if non of them are essential in this case they are known as cyclic PI.

for example:

A \ BC	00	01	11	10
0	1	1		
1	1		1	1

R P Q

P = {0, 4}
Q = {6, 7} → Essential implicant
R = {4, 6} → Redundant implicant

Minimal cover:

= Essential PI + (PI not covered in EI)

Procedure:

Step 1:

→ Construct Prime implicant
- 2 adj minterms → Pair (Eliminating 1 literal)
- 4 adj minterms → Quad (Eliminating 2 literal)
- 8 adj minterms → Octal (Eliminating 3 literal)

Step 2:

→ Identify essential prime implicants

Step 3:

→ Construct minimal expression

Example's

(1)

A \ BC	00	01	11	10
0	0	1	0	0
1	1	0	1	1

(6,7), (4,6), (1)

$= AB + A\bar{C} + \bar{A}\bar{B}C$

ii)

A \ BC	00	01	11	10
0			1	
1	1		1	1

(4,6) (6,7) (3,7)

(6,7) not essential

$= A\bar{C} + BC$

iii)

A \ BC	00	01	11	10
0		1	1	1
1		1	1	

$= C + \bar{A}$

iv)

BC				
	1	1	1	1
	1			1

$= \bar{A} + \bar{C}$

v)

AB \ CD				
	1			1
	1			1

$= \bar{B}\bar{D}$

AB \ CD	00	01	11	10
00	m_0 0	m_1 1	m_3 3	m_2 2
01	m_4 4	m_5 5	m_7 7	m_6 6
11	m_{12} 12	13	m_{15} 15	m_{14} 14
10	m_8 8	m_9 9	m_{11} 11	m_{10} 10

Simplify the following boolean expression.

$F = \Sigma m(0,1,2,4,8,9,10)$

AB \ CD	00	01	11	10
00	1	1		1
01	1			
11				
10	1	1		1

$= \bar{A}\bar{B}\bar{C} + \bar{B}\bar{C} + \bar{B}C\bar{D}$ Ans

Simplify the following boolean expression.

$F(A,B,C) = \Sigma m(0,1,2,3,4,5,6)$

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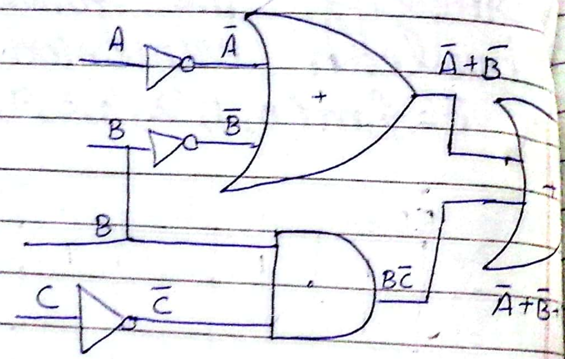
A \ BC	00	01	11	10
0	1	1	1	1
1	1	1		1

$= \bar{B} + \bar{A} + B\bar{C}$



Simlify the following boolean expression into (SOP) term.

$F(x,y,z) = xy + y\bar{z} + xyz + \bar{x}\bar{y}\bar{z} + \bar{x}$

$= xy(z+\bar{z}) + (x+\bar{x})y\bar{z} + xyz + \bar{x}\bar{y}\bar{z} + \bar{x}(y+\bar{y})(z+\bar{z})$

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$= xyz + xy\bar{z} + xy\bar{z} + \bar{x}y\bar{z} + xyz + \bar{x}\bar{y}\bar{z} + \bar{x}(yz + y\bar{z} + \bar{y}z + \bar{y}\bar{z})$

$= xyz + xy\bar{z} + xy\bar{z} + \bar{x}y\bar{z} + xyz + \bar{x}\bar{y}\bar{z} + \bar{x}yz + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$

$= xyz + \bar{x}\bar{y}\bar{z} + xy\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}yz$

$= xy(z+\bar{z}) + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}yz$

$= xy + \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + \bar{x}\bar{y}z + \bar{x}yz$

$= xy + \bar{x}\bar{y}(z+\bar{z}) + \bar{x}y\bar{z} + \bar{x}yz$

$= xy + \bar{x}\bar{y} + \bar{x}y(z+\bar{z})$

$= xy + \bar{x}\bar{y} + \bar{x}y$

$= xy + \bar{x}(y+\bar{y})$

$= xy + \bar{x}$

x \ yz	00	01	11	10
0	1	1	1	1
1			1	1

$\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}yz + \bar{x}y\bar{z}$: $\bar{x}$ is common

$\bar{x}yz$ $\quad \bar{x}y\bar{z}$; y is common

xyz $\quad xy\bar{z}$

Use karnaugh map to minimize the following SSOP expression

$\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} + AB\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D}$

AB \ CD	00	01	11	10
00	1		1	1
01	1			1
11	1			1
10	1		1	1

first term must be expand $\bar{B}\bar{C}\bar{D}$

$\bar{B}\bar{C}\bar{D}(A+\bar{A})$

$A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$

$= \bar{D} + \bar{B}C$

K-map Maxterm Simplification (POS)

Maxterm:

→ They are the sum of i/p such that total value of sum = 0.

Simplify the boolean expression in POS term using four variable K-Map.

$F(A,B,C,D) = \pi(1,3,4,5,6,7,12,13)$

AB \ CD	00	01	11	10
00		0	0	
01	0	0	0	0
11	0	0		
10				

$= \overline{\bar{A}B + \bar{A}D + B\bar{C}}$

$= \overline{\bar{A}B} + \overline{\bar{A}D} + \overline{B\bar{C}}$

$= (A+\bar{B})(A+\bar{D})(\bar{B}+C)$

Simplify the boolean expression in POS term using four variable K-map.

$F(A,B,C) = \pi M(0,1,3)$

A \ BC	00	01	11	10
0	0	0	0	
1				

$= \bar{A}\bar{B} + \bar{A}C$

Apply De morgan's law

$=$

$= (A+B)(A+\bar{C})$

Simplify the following boolean expression

$F(A,B,C,D) = \pi(3,4,6,7,11,12,13,14,15)$

AB \ CD	00	01	11	10
00			0	
01	0		0	0
11	0	0	0	0
10			0	

$= \overline{(CD) + (B\bar{D}) + (AB)}$

$= (\bar{C}+\bar{D}) \cdot (\bar{B}+D) \cdot (\bar{A}+\bar{B})$

Simplify the following boolean expression

$F(X,Y,Z) = \pi M(1,2,3,4,9,12)$

AB \ CD	00	01	11	10
00		0	0	0
01	0			
11	0			
10		0		

$= \overline{(\bar{A}\bar{B}D) + (\bar{A}\bar{B}C) + (B\bar{C}\bar{D}) + (\bar{B}\bar{C}D)}$

Apply de morgan's law

$= (A+B+\bar{D})(A+B+\bar{C})(\bar{B}+C+D)$

Simplify the following boolean expression

$Y = \pi M(1, 3, 4, 5, 6, 7, 12, 13)$

AB \ CD	00	01	11	10
00		0	0	
01	0	0	0	0
11	0	0		
10				

$= \bar{A}B + \bar{A}D + B\bar{C}$

Apply De morgan law

$= \overline{(\bar{A}B) + \bar{A}D + B\bar{C}}$

$= (\overline{\bar{A}B})(\overline{\bar{A}D})(\overline{B\bar{C}})$

$= (A+\bar{B})(A+\bar{D})(\bar{B}+C)$

Use a karnaugh map to minimize the following standard pos expression.

$(A+B+C)(A+B+\bar{C})(A+\bar{B}+C)(A+\bar{B}+\bar{C})(\bar{A}+\bar{B}+C)$

A \ BC	00	01	11	10
0	0	0		
1	0	0		0

$\bar{B} + A\bar{C} = B(\bar{A}+C)$

Don't Care Conditions

→ Don't care conditions are part of function specification.

$$f = \Sigma m(\ldots\ldots) + \Sigma d(\ldots)$$

$$f = \pi M(\ldots\ldots) + \pi D(\ldots\ldots)$$

→ They can be used for both sum-of-product and product of sum forms of functions.

→ A Don't care cell as 1 or 0 or we also ignore that cell.

→ A Don't care cell can be represented by a cross (X) in K-maps representing an invalid combination.

→ Don't care help us to form larger group of cells.

Simplify the following boolean expression Using Don't care condition.

$F = \Sigma m(1,3,7) + \Sigma d(0,5)$

A \ BC	00	01	11	10
0	X 0	1 1	1 3	2
1	4	X 5	1 7	6

$F = C$

$F(A,B,C,D) = \Sigma m(1,3,7,11,15) + \Sigma d(0,2,5)$

AB \ CD				
	0	1 1	1 3	X 2
	4	X 5	1 7	6
	12	13	1 15	14
	8	9	1 11	10

$F = CD + \bar{A}D$

Simplify the following boolean expression $\Sigma m(1,3,5,7,9) + \Sigma d(6,12,13)$

AB \ CD	00	01	11	10
00		1	1	
01		1	1	
11	X	X		
10		1		

$= \bar{C}D + \bar{A}D$

Simplify the following boolean expression $F(P,Q,R,S) = \Pi M(0,2,3,4,5,6,8,10,11,12,15) + \Sigma D\,\Pi M(1,7,13)$

PQ \ RS	00	01	11	10
00	0	X	0	0
01	0	0	X	0
11	0	X	0	
10	0		0	0

$= \bar{P} + \bar{R}\bar{S} + RS + \bar{Q}R$

$= P(R+S)\cdot(\bar{R}+\bar{S})(\bar{Q}+\bar{R})$